

Principal Component Analysis

Identifying Key Variance Drivers in Hospital Patient Data

Research Question

Which factors from the medical dataset cause the greatest variance in other dataset variables?

Analysis Goal

One goal of this data analysis will be to determine a recommended number of principal components for future data analysis.

Method Justification

Principal component analysis evaluates each selected variable to assess the eigen value, representing the amount of change in values across the whole database caused per variable. Eigen values above one indicate a principal component should be included in the reduced dataset.

The expected outcome of this analysis is a recommended number of principal components that represent the dataset, along with numeric measurement of the variance caused by each component.

PCA assumes that relationships between variables are generally linear.

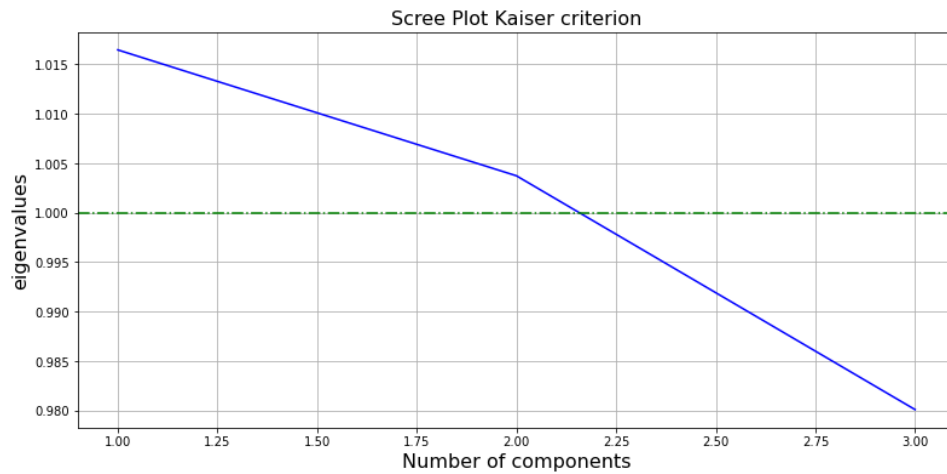
Variables

The selected variables will be the continuous variables in the medical dataset: income, vitD_levels, and Initial_days.

Principal Components Matrix

```
In [10]: print(loading_matrix)
          Initial_days  Income  VitD_levels
PC0      -0.455065    0.741561   -0.492953
PC1       0.726339   -0.011124   -0.687246
PC2       0.515119    0.670793    0.533563
```

Scree Plot & Kaiser Criterion



The Kaiser criterion recommends two principal components.

Component Variance

```
{'PC0': 1, 'PC1': 2}
Variance of first 2 PCs: [1.01646898 1.00373705]
total variance: 0.6733346683809162
```

PC0 Variance: 1.016

PC1 Variance: 1.003

The total variance captured by the two selected principal components is 0.673.

Results

Principal component analysis has been implemented, reducing the dataset from three continuous variables to two principal components. A loading matrix was built, showing how these dataset variables correlate with one another. Using the Kaiser criterion, two principal components were selected as they showed eigenvalues greater than one.

Finally, the variance of the final two components has been analyzed along with calculating the total variance of those two combined, equaling 67.3%.